6. Public Wi-fi

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6.1 Safe use of public WI-FI

Public Wi-Fi is available in many places such as universities, libraries, cafés, airports, hotels and shopping centres. It provides a convenient way for people to access the internet when they are away from home. However, public Wi-Fi can also create security and privacy risks if it is not used carefully. Users should understand these risks and take appropriate precautions to protect their personal information, accounts and devices. (FTC, Are public wifi networks safe what you need to know, 2023)

One of the main risks of public Wi-Fi is that not every network is trustworthy. Some networks may be poorly secured, while others could be deliberately created by attackers to trick people into connecting. For example, someone could create a Wi-Fi network with a name similar to a legitimate café or university network. A user may connect without checking whether it is the official network. Once connected, their internet activity could potentially be exposed to an attacker.

Another risk is the theft of personal information. When using an unsecured network, sensitive information such as usernames, passwords or other data may be at greater risk, particularly when using websites that do not properly protect communications. This is why users should avoid entering highly sensitive information, such as online banking details, while connected to an unfamiliar public network. If sensitive tasks are necessary, it is safer to use a trusted mobile connection or another secure network. (FTC, FTC Provides Tips for Using VPN Apps, 2018)

6.2 VPN protection and HTTPS

Users should also make sure that websites use HTTPS. HTTPS encrypts the communication between a user's browser and the website, making it more difficult for other people on the network to read the information being exchanged. Most modern browsers display a padlock or similar security indicator when a connection is protected by HTTPS. However, users should still be careful because HTTPS does not automatically mean that a website itself is trustworthy.

A VPN (Virtual Private Network) can provide an additional layer of protection when using public Wi-Fi. A VPN encrypts traffic between the device and the VPN service, which can make it harder for other people on the same Wi-Fi network to monitor the user's network activity. However, users should choose a reputable VPN provider and understand that a VPN does not protect against every type of threat. For example, it does not prevent someone from being tricked by a phishing website or downloading malicious software. (NCSC, Virtual Private Networks, 2021)

It is also important to keep devices up to date. Operating system and application updates often contain security fixes for vulnerabilities that attackers could otherwise exploit. Before using public Wi-Fi, users should make sure their computer, phone or tablet has current security updates. Antivirus or endpoint security software can also provide additional protection against malicious files and programs. (NCSC, Keeping devices and software up to date, 2021)

Another useful precaution is to disable automatic Wi-Fi connections. If a device automatically connects to networks, it has previously used, it may connect without the user realising it. Turning off automatic connection settings gives the user more control over which networks their device joins. Users should also avoid connecting to networks with suspicious names or networks that do not require any authentication when an official network is available. (FTC, Are public wifi networks safe what you need to know, 2023)

File and device sharing should also be disabled when using an unfamiliar network. Features that allow other devices to discover or connect to a computer can increase the amount of information exposed to other people on the same network. Users should check their device's network and sharing settings and use appropriate security settings, particularly when connecting in public places.

6.3 Practical Precautions

Finally, users should remember to log out of important accounts after using a shared or public computer and avoid saving passwords on devices that they do not own. Multi-factor authentication (MFA) should also be enabled on important accounts whenever possible. MFA provides an additional security step, meaning that a stolen password alone may not be enough to access an account. (NCSC, Why MFA matters, 2024b)

In conclusion, public Wi-Fi is useful and convenient, but it should not automatically be considered safe. Users can reduce the risks by connecting only to legitimate networks, using HTTPS, considering a reputable VPN, keeping devices updated, disabling

automatic connections and file sharing, avoiding sensitive activities on untrusted networks, and using strong authentication. By following these practices, people can use public Wi-Fi more safely while reducing the likelihood of their personal information or accounts being compromised.